Dhirubhai Ambani University

Course Title: Analog and Mixed Signal IC

Course Code: EL435

Course Credit: 2-0-4-4

Prerequisite: Basic understanding of VLSI Design

Course Placement: EL435 is a B.Tech. EVD specialization course for B.Tech. VI Sem (EVD) students.

Course Content:

The course aims to impart VLSI, MOSFET concepts, advanced circuit design concepts with Device and Design Methodology, Inversion Coefficient and Gm/ID Design Methodology, Continuous/Discrete ~ time Amplifiers, OTAs and OPAMP, Comparators, Analog ~ to ~ digital / Digital ~ to ~ Analog Circuits.

Books:

1. CMOS Digital Integrated Circuits-Analysis & Design by S.M. Kang & Y. Leblibici, TMH.
2. B. Razavi ~ Design of Analog CMOS Integrated Circuits (2nd Edition, Mc Graw Hill, 2017).
3. W. Sansen ~ Analog Design Essentials (Springer, 2013).
4. A. Sedra & K. Smith ~ Microelectronic Circuits (7th Edition, Oxford Univ. Press, 2015).
5. P. R. Gray, P. J. Hurst, S. H. Lewis & R. G. Meyer ~ Analysis and Design of Analog Integrated Circuits (5th Edition., Wiley, 2009).

Assessment Method: In-Semester Examinations and a laboratory Examination will be conducted during the course.

Grading Policy:

In-Sem 1:	**20%**
In-Sem 2:	**20%**
End-Sem:	**20%**
Lab:	**30%**
Class Performance, Quiz, Project:	**10%**

Course Outcomes:

After successful completion of this course,

1. The students shall be able to comprehend transistors, basic VLSI design flow, amplifiers, converters.
2. Use of EDA tools and their efficient execution to implement the system design.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X							X